DES Pune University, Pune
School of Engineering and Technology
Department of Computer Engineering and Technology
Program: B. Tech in Computer Science and Engineering

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Subject: Python Programming		
Assignment No.: 18		
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Lab Assignment: 18

Aim:
Write a program to create their own module for a specific function.

Objective:
To understand how to create and use a user-defined module in Python within Jupyter Notebook.

Theory:
A **module** in Python is a file containing functions, variables, or classes that can be reused in other programs.

- Modules improve code reusability and organization.
- A user-defined module is created using a .py file.
- The module is imported using the import keyword.
- Functions are accessed using module_name.function_name().

In Jupyter Notebook, the %%writefile magic command is used to create a module file directly from a notebook cell.

Algorithm:
Part 1: Creating the Module

1. Start the program.
2. Use %%writefile to create a file named mymath.py.
3. Define functions:
    - add(a, b)
    - subtract(a, b)
    - multiply(a, b)
    - divide(a, b)
4. Save the file.

Part 2: Using the Module

1. Import the created module.
2. Enter two numbers.
3. Call each function from the module.
4. Display the results.
5. Stop the program.

```
Pseudo Code:
START
CREATE module mymath
    DEFINE add(a, b)
        RETURN a + b

    DEFINE subtract(a, b)
        RETURN a - b

    DEFINE multiply(a, b)
        RETURN a * b

    DEFINE divide(a, b)
        IF b != 0
            RETURN a / b
        ELSE
            RETURN error message
END MODULE

IMPORT mymath
READ num1, num2
PRINT add(num1, num2)
PRINT subtract(num1, num2)
PRINT multiply(num1, num2)
PRINT divide(num1, num2)
STOP
```

Code:

```python
%%writefile mymath.py

def add(a, b):
    return a + b

def subtract(a, b):
    return a - b

def multiply(a, b):
    return a * b

def divide(a, b):
    if b != 0:
        return a / b
    else:
        return "Division by zero is not allowed"
```

```
import mymath
```

```
num1 = float(input("Enter first number: "))
num2 = float(input("Enter second number: "))

print("Addition:", mymath.add(num1, num2))
print("Subtraction:", mymath.subtract(num1, num2))
print("Multiplication:", mymath.multiply(num1, num2))
print("Division:", mymath.divide(num1, num2))
```

Output:

```
Enter first number:  10
Enter second number:  5
Addition: 15.0
Subtraction: 5.0
Multiplication: 50.0
Division: 2.0
```

Conclusion:

Thus, a user-defined module was successfully created and used in Jupyter Notebook. The program performed addition, subtraction, multiplication, and division operations correctly, demonstrating modular programming and code reusability in Python.